

Conditional statements

-if,if elif else,nested if

```
In [19]: if True:  
         print("Data Science")
```

Data Science

```
In [20]: if False:  
         print("Data Science")
```

```
In [21]: #if else  
         if True:  
             print('Data Science')  
         else:  
             print('by by')
```

Data Science

```
In [22]: if False:  
         print('Data Science')  
         else:  
             print('by by')
```

by by

```
In [23]: age = 24  
         if age<13:  
             print("you are a child .")  
         elif age<20:  
             print("you are a teenager .")  
         else:  
             print("you are a adult.")
```

you are a adult.

```
In [ ]: x=4  
         r=x % 2  
         if r==0:  
             print("x is even")  
         else:  
             print("x is odd")
```

```
In [ ]: x = 6  
         r = x % 2  
  
         if r == 0:  
             print('Even number')  
             if x>5:  
                 print('greater number')  
  
         else:  
             print('Odd Number')
```

```
In [26]: x = 2  
         r = x % 2
```

```
if r == 0:
    print('Even number')

    if x>5:
        print('greater number')
    else:
        print('not greater')

else:
    print('Odd Number')
```

Even number
not greater

```
In [28]: year = int(input("Enter a year: "))

if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
    print("Leap Year")
else:
    print("Not a Leap Year")
```

Not a Leap Year

```
In [45]: # elif it wont check till the block once you find the output it wont go to next

x = 4

if(x == 1):
    print('one')
elif(x == 2):
    print('Two')
elif(x == 3):
    print('Three')
elif(x == 4):
    print('four')
```

four

```
In [46]: # elif it wont check till the block once you find the output it wont go to next

x = 7

if(x == 1):
    print('one')
elif(x == 2):
    print('Two')
elif(x == 3):
    print('Three')
elif(x == 4):
    print('four')
```

```
In [47]: x = 7

if(x == 1):
    print('one')

elif(x == 2):
    print('Two')
elif(x == 3):
    print('Three')
elif(x == 4):
```

```

    print('four')

else:
    print('number not found')

```

number not found

loops

```

In [32]: print("Data Science")
         print("Data Science")
         print("Data Science")
         print("Data Science")
         print("Data Science")

```

Data Science
Data Science
Data Science
Data Science
Data Science

loops divide in to two types while loop and for loop

```

In [33]: #while loop
         i=1 #initialization
         while i<=5: #condition
             print("data science",i)
             i=i+1 #increament

```

data science 1
data science 2
data science 3
data science 4
data science 5

```

In [34]: i=5 #initialization
         while i>=1: #condition
             print("data science",i)
             i=i-1 #increament

```

data science 5
data science 4
data science 3
data science 2
data science 1

```

In [1]: i = 1

         while i<=5:
             print('data science') # when we mention end then new line will not create
             j = 1
             while j<=4:
                 print('technology')
                 j = j + 1

             i = i + 1
             print()

         # the output which we got is very lengty but how to make them one line Lets

```

data science
 technology
 technology
 technology
 technology

data science
 technology
 technology
 technology
 technology

data science
 technology
 technology
 technology
 technology

data science
 technology
 technology
 technology
 technology

data science
 technology
 technology
 technology
 technology

```
In [2]: i = 1
while i<=5:
    print(' datascience', end = "") # when we mention end then new line will not
    j = 1
    while j<=4:
        print(' technology', end="")
        j = j + 1

    i = i + 1
    print()
```

datascience technology technology technology technology
 datascience technology technology technology technology
 datascience technology technology technology technology
 datascience technology technology technology technology
 datascience technology technology technology technology

```
In [2]: i=1
while i<=2:
    j=0
    while j<=2:
        print(i*j,end="")
        j+=1
    print()
    i+=1
```

0
1
2
0
2
4

```
In [4]: i=1
while i<=4:
    j=0
    while j<=3:
        print(i*j,end="")
        j+=1
        print()
    i+=1
```

0
1
2
3
0
2
4
6
0
3
6
9
0
4
8
12

```
In [3]: for i in range(1,11):
        if i==6:
            break
        print(i)
```

1
2
3
4
5

```
In [5]: for i in range(1,11):
        if i==6:
            continue
        print(i)
```

1
2
3
4
5
7
8
9
10

```
In [9]: for i in range(1,11):
```

```
Cell In[9], line 1
    for i in range(1,11):
                        ^
SyntaxError: incomplete input
```

```
In [10]: for i in range(1,11): #pass will pass the error
        pass
```

```
In [2]: i=1
        while i<=4:
            j=0
            while j<=3:
                print(i*j,end='')
                j+=1
            print()
            i+=1
```

```
0123
0246
0369
04812
```

```
In [4]: name1=[1,2,5,'hello']
        for i in name1:
            print(i)
```

```
1
2
5
hello
```

```
In [5]: for i in range(5):
        print(i)
```

```
0
1
2
3
4
```

```
In [6]: for i in range (1,51):
        if i % 5 == 0:
            print(i)
```

```
5
10
15
20
25
30
35
40
45
50
```

```
In [7]: #Write a code user asking from the vending machine for a chocolate
x=int(input("How many chocolates you want:"))
i=1
while i<=x:
```

```
print('chocolate')  
i+=1
```

chocolate
chocolate
chocolate
chocolate
chocolate

```
In [8]: #what if the wending machine has only 5 chocolates  
ave=5  
x=int(input("How many chocolates you want:"))  
i=1  
while i<=x:  
    print('chocolate')  
    i+=1
```

chocolate
chocolate
chocolate
chocolate
chocolate
chocolate
chocolate

```
In [9]: ava_choco=5  
x = int(input("How many chocolates you want:"))  
i=1  
while i<= x:  
    if i>ava_choco:  
        break  
    print('chocolate')  
    i +=1  
print('Bye for now')
```

chocolate
chocolate
chocolate
chocolate
chocolate
Bye for now

```
In [12]: for i in range(1,51):  
    if i% 3==0:  
        print(i)
```

3
6
9
12
15
18
21
24
27
30
33
36
39
42
45
48

```
In [18]: for i in range (1,51):  
         if i % 3==0 or i% 5==0:  
             continue  
         print(i)
```

1
2
4
7
8
11
13
14
16
17
19
22
23
26
28
29
31
32
34
37
38
41
43
44
46
47
49

For Else

```
In [19]: nums=[12,15,18,21,26,30,40]  
         for num in nums:  
             if num % 5 == 0:  
                 print(num)
```


15
30
40

```
In [22]: nums=[12,15,18,21,26,30,40]
         for num in nums:
             if num % 5 == 0:
                 print(num)
                 break
```

15

```
In [24]: nums=[7,14,18,21,23,27,29]
         for num in nums:
             if num % 5 ==0:
                 print(num)
                 break
         else:
             print("Number is not divisible")
```

Number is not divisible

print pattern in python

```
In [25]: print("#####")
         print("#####")
         print("#####")
         print("#####")
```


#####

```
In [26]: for i in range (1,5):
         i =i+1
         print('####')
```


####

```
In [27]: for i in range (1,10):
         i=i+1
         print('# # # #')
```


#

```
In [29]: for i in range(1,5):
         if i<=5:
             print('# # # #')
```

```
# # # #
# # # #
# # # #
# # # #
```

```
In [31]: for j in range(4):
         print('#')
```

```
#
#
#
#
```

```
In [32]: for j in range(4):
         print('#',end="")
```

```
####
```

```
In [34]: for j in range(4):
         print('#', end= " ")
         print()
         for j in range(4):
             print('#',end=" ")
```

```
# # # #
# # # #
```

```
In [39]: for i in range (4):
         for j in range(i+1):
             print('#',end = ' ')
         print()
```

```
#
# #
# # #
# # # #
```

How to check the number is prime

```
In [44]: num=int(input("enter a number:"))
         for i in range(2,num):
             if num%i==0:
                 print("number is not prime")
                 break
             else:
                 print("number is prime")
```

```
number is prime
```

```
In [ ]:
```